

NEERAJ SOLANKI

nsola5@uic.edu — linkedin.com/in/neeraj-solanki — Google Scholar — solankineeraj03.github.io — Chicago, Illinois

Work Experience

AI/ML Research Intern

May 2026 to Aug 2026

Qualcomm, Cloud AI Division

Santa Clara, USA

- Built reusable inference-onboarding and benchmarking workflows for GPT-2, MiniVLA, and custom-attention models on Qualcomm Cloud AI 100 Ultra; achieved up to $7.05\times$ higher throughput while preserving numerical accuracy.
- Reduced model-onboarding time by 41 to 52% through a reusable hardware-onboarding workflow spanning language, vision-language-action, and custom-attention workloads.
- Trained and evaluated a 547.6M-parameter DiT text-to-video model on $4\times$ NVIDIA H100 GPUs, designing the evaluation protocol (optical-flow consistency, FVD, reconstruction testing) that measured a 19.9% motion-consistency and 12.3% FVD improvement over the baseline attention.

Machine Learning Research Assistant

Aug 2024 to Present

University of Illinois Chicago

Chicago, USA

- Designed adaptive inference using early-exit and precision-aware execution, achieving $2\times$ throughput improvement while preserving perceptual accuracy under compute constraints.
- Implemented lightweight neural architectures for real-time signal classification, reducing memory footprint by $100\times$ while maintaining perceptually relevant performance metrics.
- Scaled multimodal learning experiments across 30+ distributed clients, reducing communication overhead by 80% for pipelines integrating temporal biosignals and perceptual representations.

Machine Learning Engineer

Jul 2022 to Jul 2024

Deloitte USI

Mumbai, India

- Engineered end-to-end production ML pipelines spanning data preprocessing, feature extraction, model training, serving, and runtime optimization; reduced processing latency by 40% across production inference workloads.
- Built and trained sequence-aware neural models for 5M+ monthly data streams, refining training objectives and evaluation pipelines to improve anomaly detection precision by 20%.

Technical Skills

Languages: Python, C++, Java, SQL.

Inference Stacks: vLLM, PyTorch, OpenAI-compatible serving, continuous batching, KV/prefix caching, quantization (AWQ, 32/8/4-bit).

Benchmarking & Experiments: TTFT/TPOT/p95 latency, throughput and quality-regression benchmarks, prefill vs decode characterization, roofline analysis, custom GPU telemetry, controlled A/B design, confound isolation, CI regression gates.

Infrastructure: Docker, AWS, GCP, GitHub Actions, Linux, REST APIs.

Education

University of Illinois Chicago

Aug 2024 to May 2026

Master of Science, Computer Science

GPA: 3.63/4

MIT World Peace University, Pune

Jun 2018 to Jun 2022

Bachelor of Technology, Computer Engineering

GPA: 3.6/4

Projects

InferScope: LLM Inference Benchmarking & Parity Harness | Python, vLLM, GPU Telemetry, Docker, CI [GitHub](#)

- Open-source harness benchmarking latency (TTFT/TPOT/p95), throughput and quality regressions across concurrency, batching and precision, with per-request tracing correlated to a custom GPU telemetry pipeline that excludes runs measured on thermally unstable hardware; 8,020 runs, 105 tests in CI.
- Quantified a quality-vs-throughput tradeoff on vLLM: throughput rose $5.2\times$ to 8-way concurrency while one fixed greedy prompt produced 1 to 20 distinct outputs. The conventional divergence-rate metric saturates near 45% and misses this; traced the cause to prefix-cache and KV-cache behavior, not the model.

AI-Assisted Radiology Reporting Platform | FastAPI, React/TypeScript, Vision-LLM, DICOM

- Built a full-stack vision-LLM platform drafting structured radiology reports on remote GPUs, then benchmarked it against Microsoft's MAIRA-2 over 210 chest X-ray studies with bootstrapped CIs: RadGraph F1 favored production (30 vs. 8) while CheXbert label agreement was statistically indistinguishable, scoping the recommendation to what the data supported.

Publications

ATM-Net: Adaptive Termination and Multi-Precision Neural Networks for Edge Intelligence (*HPCA 2025*)

[Link](#)

ARISE: Adaptive Resource-aware Intelligent SEnsing Enabling Intermittent Learning (*IEEE*)

[Link](#)

SenGuard: In-Sensor Privacy-Preserving Processing for Smart Imaging (*GLSVLSI 2025*)

[Link](#)